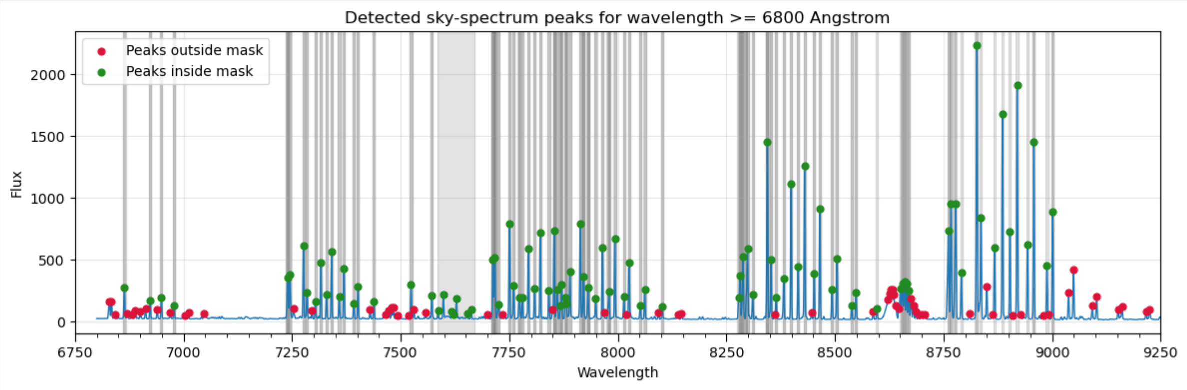
20260310 Skyline Spectral Masking

1. Creating Skyline Spectral Masking

We confirmed that we will use `DEG` = 23 for the upcoming $4800 - 9100\mathring{A}$ runs.



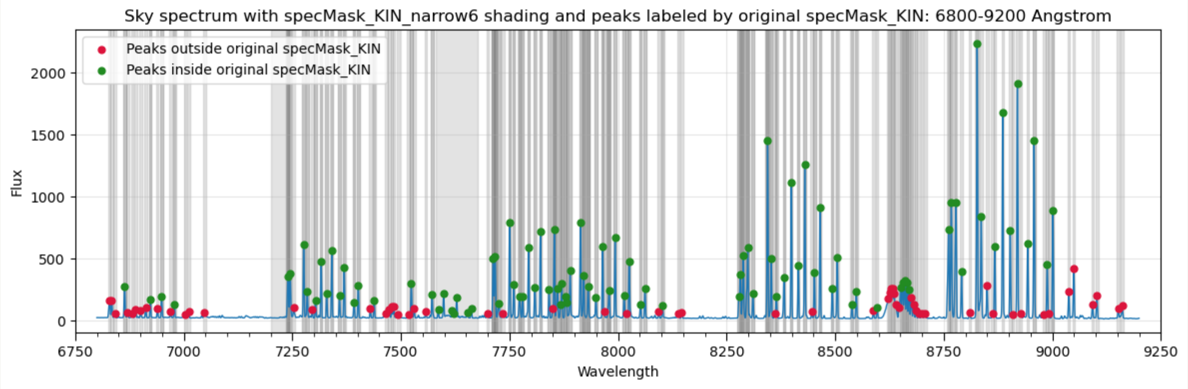
Below is my way to create the Skyline Spectral Masking for $\geq 6800\mathring{A}$:

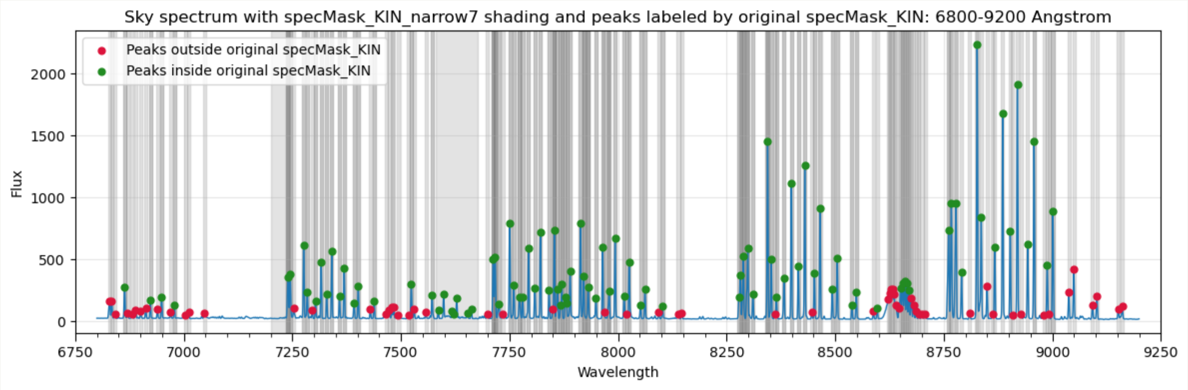
1. detect all the peaks in the sky spectrum with flux >= 50 at $6800 - 9200\mathring{A}$
2. then we distinguish two groups of peaks: green is already masked in `specMask_KIN`, while red is not masked yet.
3. For green peaks, then use the existing `specMask_KIN` to determine the masking, while for red peaks, use the detected peak wavelength as the center. As for the width, we widen all the $5\mathring{A}$ linewidth to $6\mathring{A}$ or $7\mathring{A}$ (including the exising ones in `specMask_KIN` and the newly detected ones).
4. As for the special `7628.00                84                sky` (telluric feature) in `specMask_KIN`, update its range from $84\mathring{A}$ to $96\mathring{A}$.
5. We further mask the noisy $7200 - 7240\mathring{A}$ region, which seems to minimise the bump right before [Ar III]$\lambda 7135$.
6. The original `specMask_KIN` only contains [Ar III]$\lambda 7135$ for those emission lines $\geq 6800\mathring{A}$. So I checkout the `emissionLinesPHANGS_full.config` (my modified version that turn on the fitting for all emissioin lines $\geq 6800\mathring{A}$ in original `emissionLinesPHANGS.config`), and add those gas lines, with their

widths all set to $20\AA$ as existing lines:

```
("OII", 7318.92),
("OII", 7319.99),
("OII", 7329.67),
("OII", 7330.73),
("HPa11", 8862.89),
("HPa10", 9015.3),
("SIII", 9068.6),
```

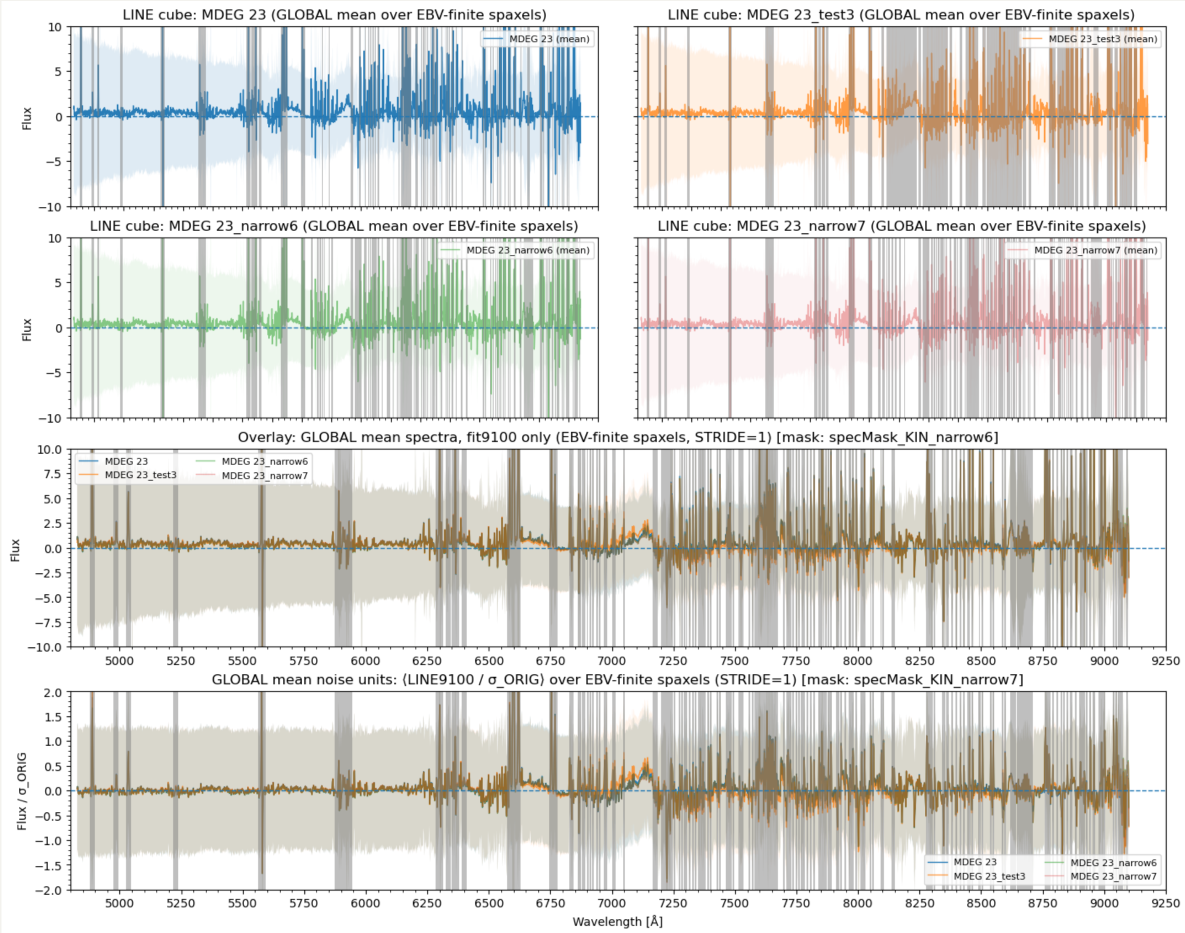
So now I have `specMask_KIN_narrow6` and `specMask_KIN_narrow7`.



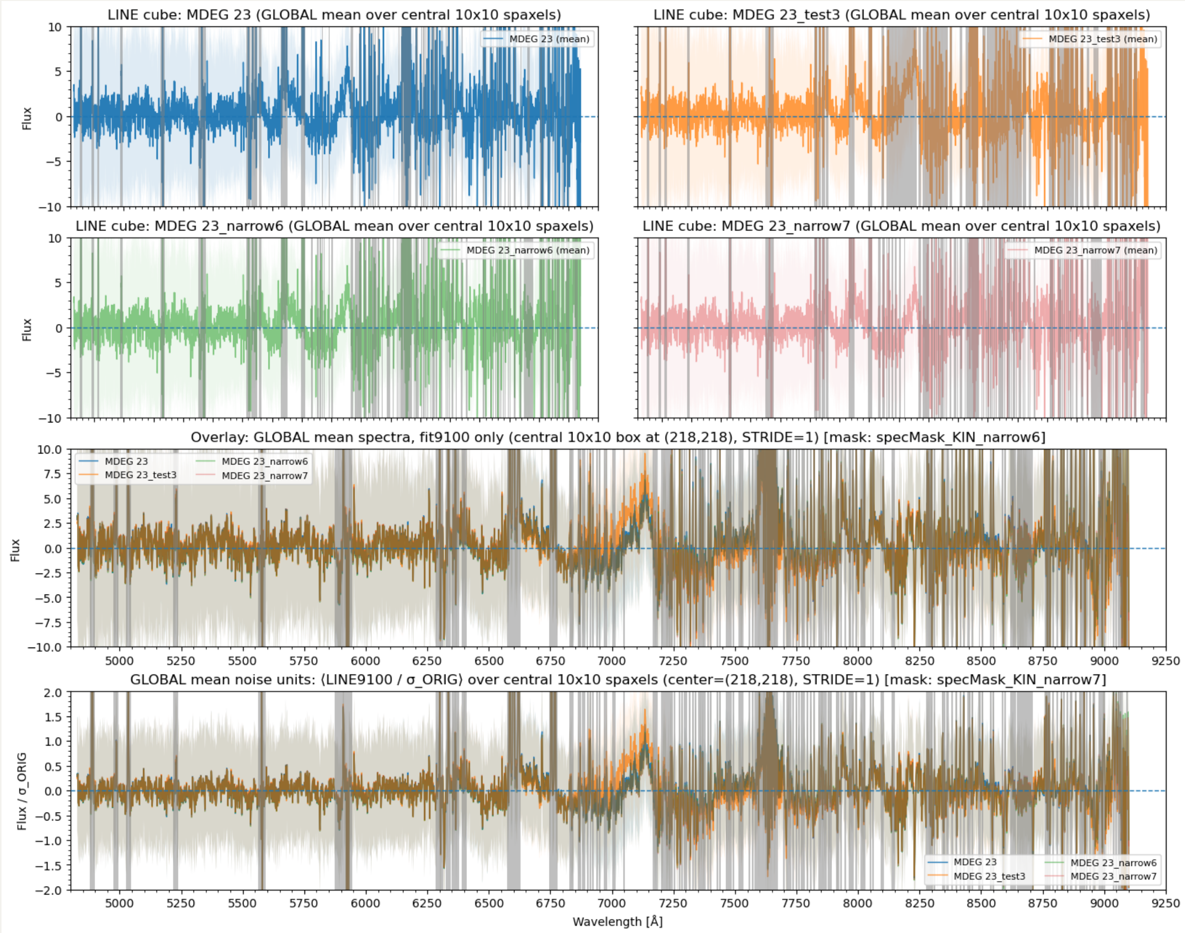


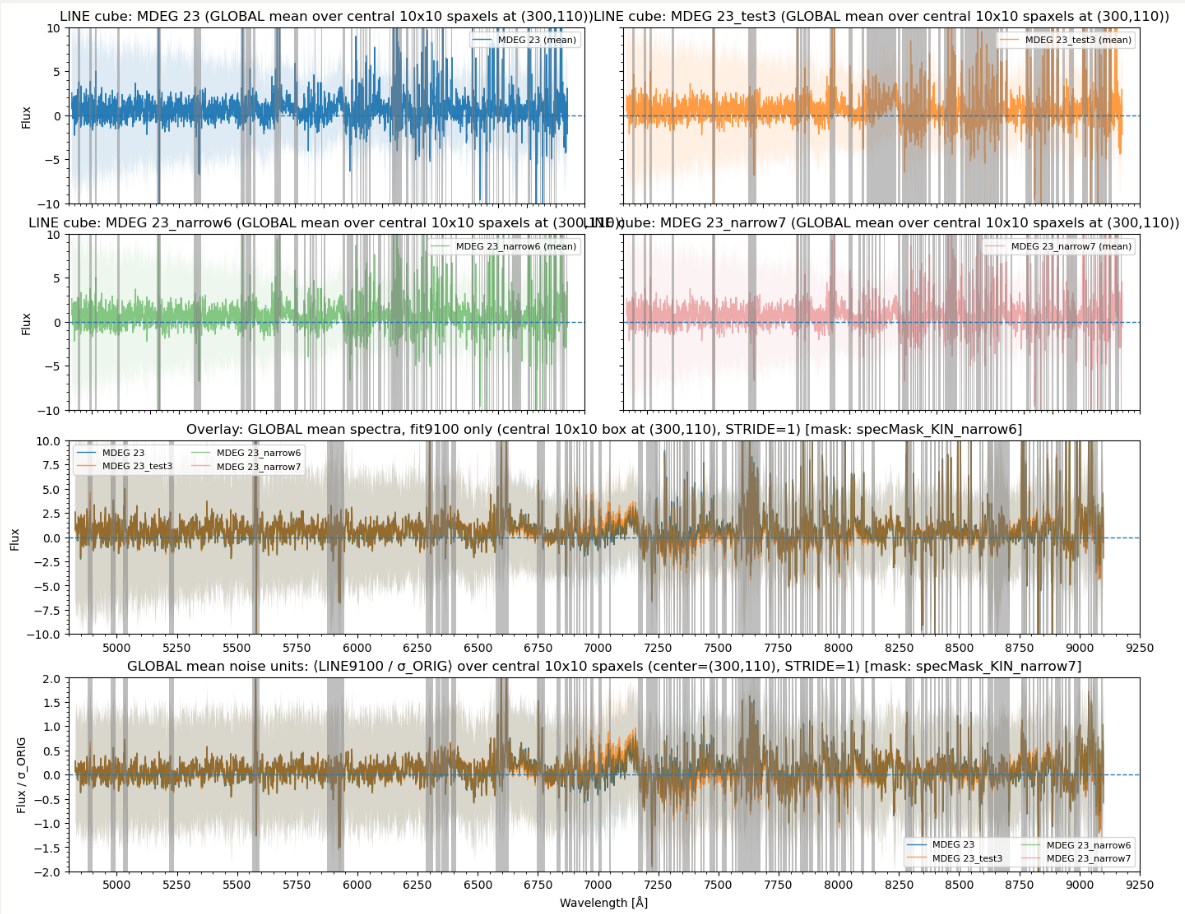
2. Checking the fitting results

Here i show the original `MDEG` =23, and the cases with `test3`, `specMask_KIN_narrow6` and `specMask_KIN_narrow7` masks of means from our spatial mask regions. I can see slight improve of reducing two bumps from original mask. It seems that `test3` provides the worst residual, so probably due to too wide masking.



Then I show the same plots from both central (high S/N) and faint (low S/N) regions.


LINE cube: MDEG 23 (GLOBAL mean over central 10x10 spaxels)
MDEG 23 (mean)
LINE cube: MDEG 23_test3 (GLOBAL mean over central 10x10 spaxels)
MDEG 23_test3 (mean)
LINE cube: MDEG 23_narrow6 (GLOBAL mean over central 10x10 spaxels)
MDEG 23_narrow6 (mean)
LINE cube: MDEG 23_narrow7 (GLOBAL mean over central 10x10 spaxels)
MDEG 23_narrow7 (mean)
Overlay: GLOBAL mean spectra, fit9100 only (central 10x10 box at (218,218), STRIDE=1) [mask: specMask_KIN_narrow6]
GLOBAL mean noise units: (LINE9100 / σ_ORIG) over central 10x10 spaxels (center=(218,218), STRIDE=1) [mask: specMask_KIN_narrow7]
MDEG 23
MDEG 23_test3
MDEG 23_narrow6
MDEG 23_narrow7
Flux
Flux / σ_ORIG
Wavelength [Å]


LINE cube: MDEG 23 (GLOBAL mean over central 10x10 spaxels at (300,110))
MDEG 23 (mean)
LINE cube: MDEG 23_test3 (GLOBAL mean over central 10x10 spaxels at (300,110))
MDEG 23_test3 (mean)
LINE cube: MDEG 23_narrow6 (GLOBAL mean over central 10x10 spaxels at (300,110))
MDEG 23_narrow6 (mean)
LINE cube: MDEG 23_narrow7 (GLOBAL mean over central 10x10 spaxels at (300,110))
MDEG 23_narrow7 (mean)
Overlay: GLOBAL mean spectra, fit9100 only (central 10x10 box at (300,110), STRIDE=1) [mask: specMask_KIN_narrow6]
GLOBAL mean noise units: (LINE9100 / σ_ORIG) over central 10x10 spaxels (center=(300,110), STRIDE=1) [mask: specMask_KIN_narrow7]
MDEG 23
MDEG 23_test3
MDEG 23_narrow6
MDEG 23_narrow7
Flux
Flux / σ_ORIG
Wavelength [Å]

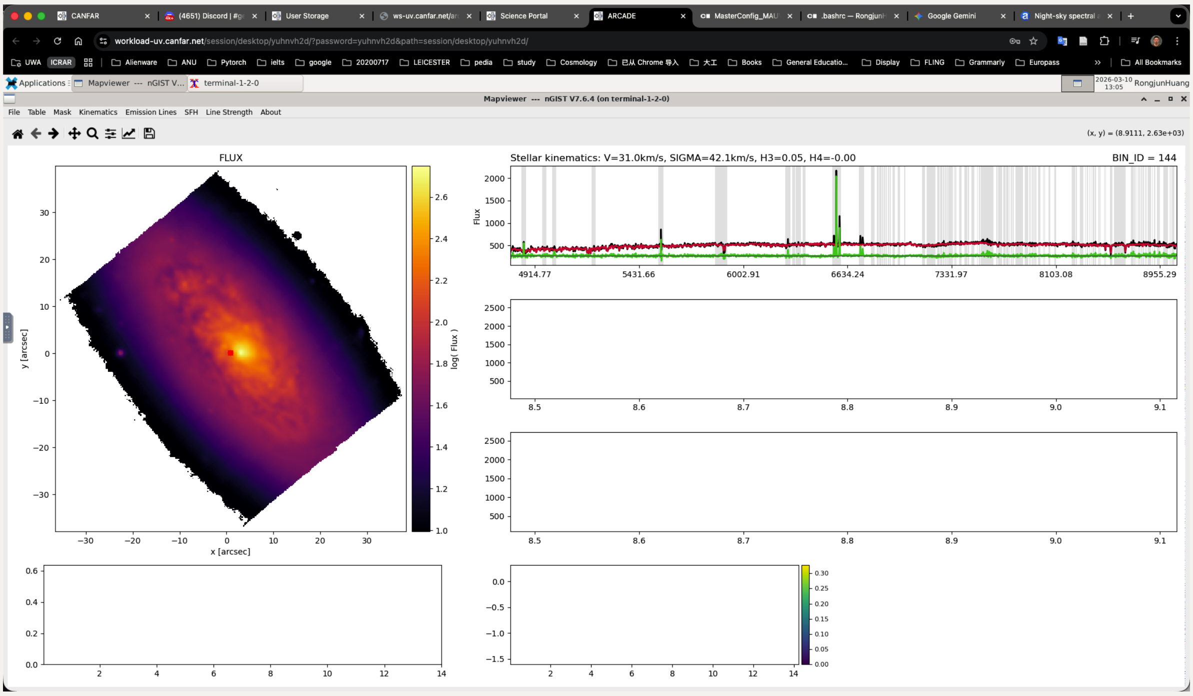
3. Checking in `Mapviewer`

Note that now the spectra are in rest frame, so the skyline masking is blue shifted here.

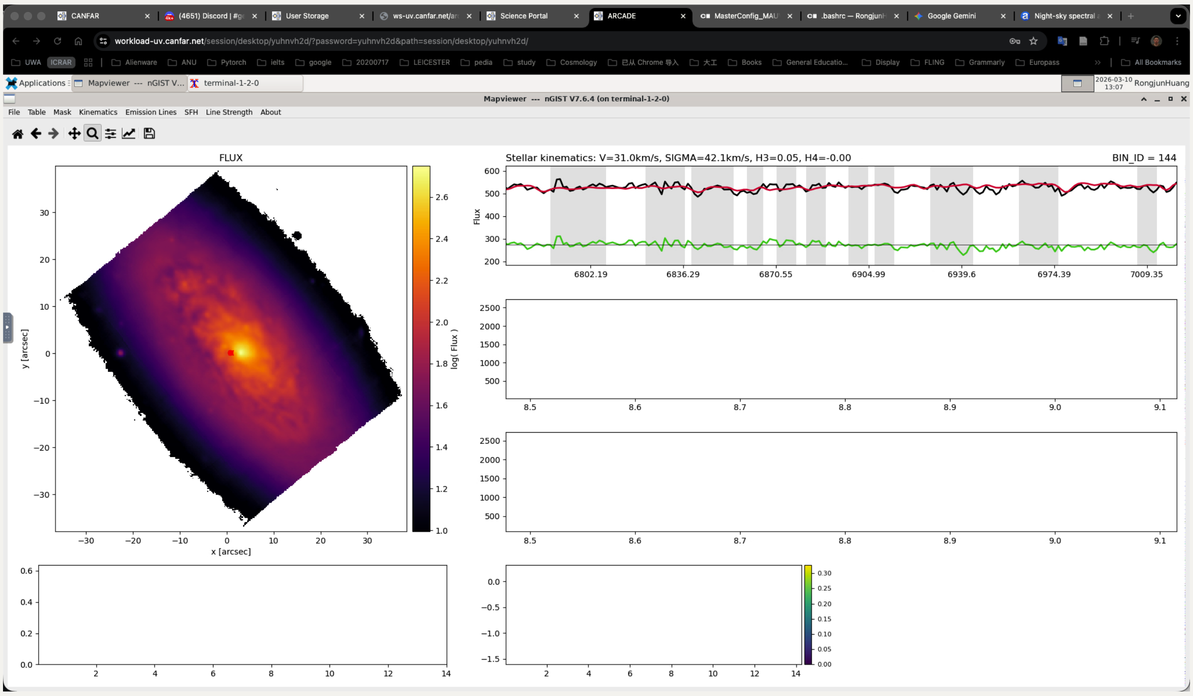
I use `specMask_KIN_narrow7` as I feel like $6\AA$ still not doesn't cover some edges (mismatch) of skylines, especially $> 7500\AA$.

3.1 Bright `Mapviewer`

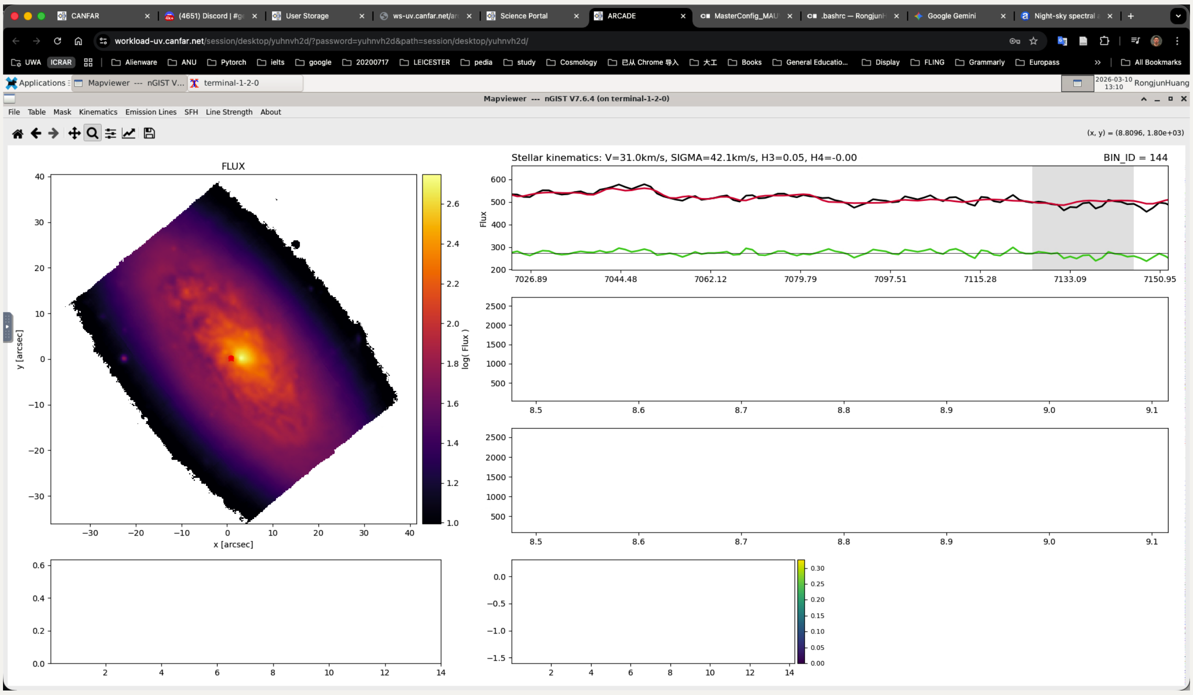
First, we still look at the central spaxel and it looks fine. The red dot roughly at (0, 0).



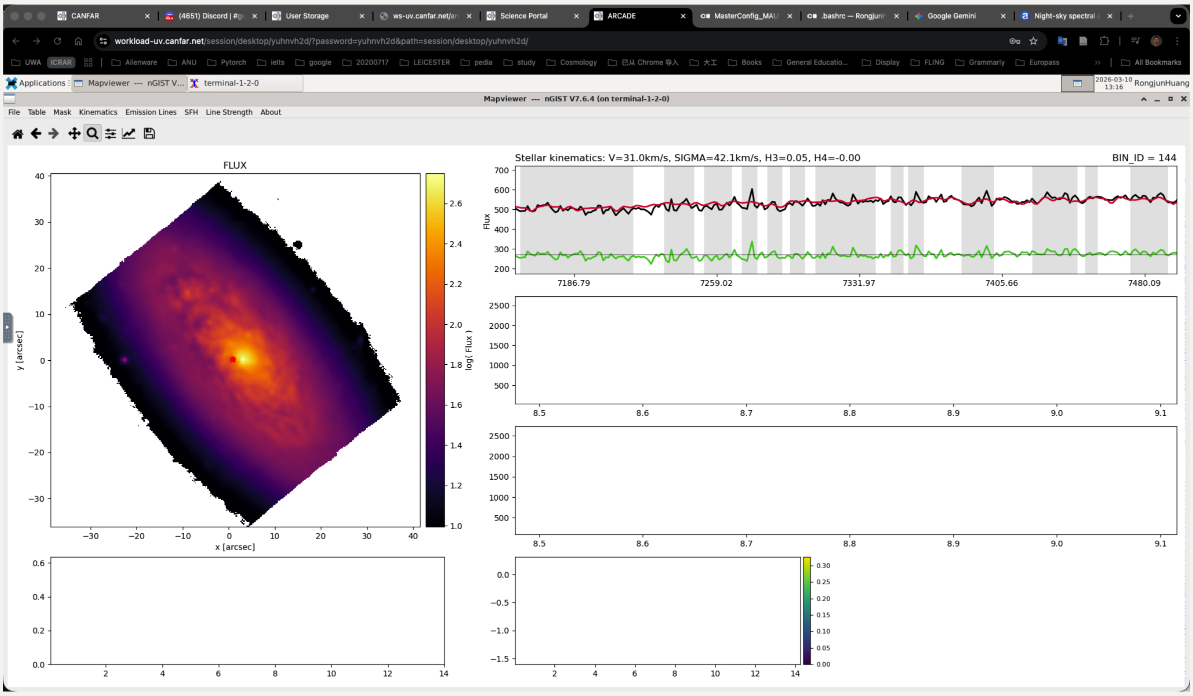
Zoom in to see $6800 \sim 7000\AA$:



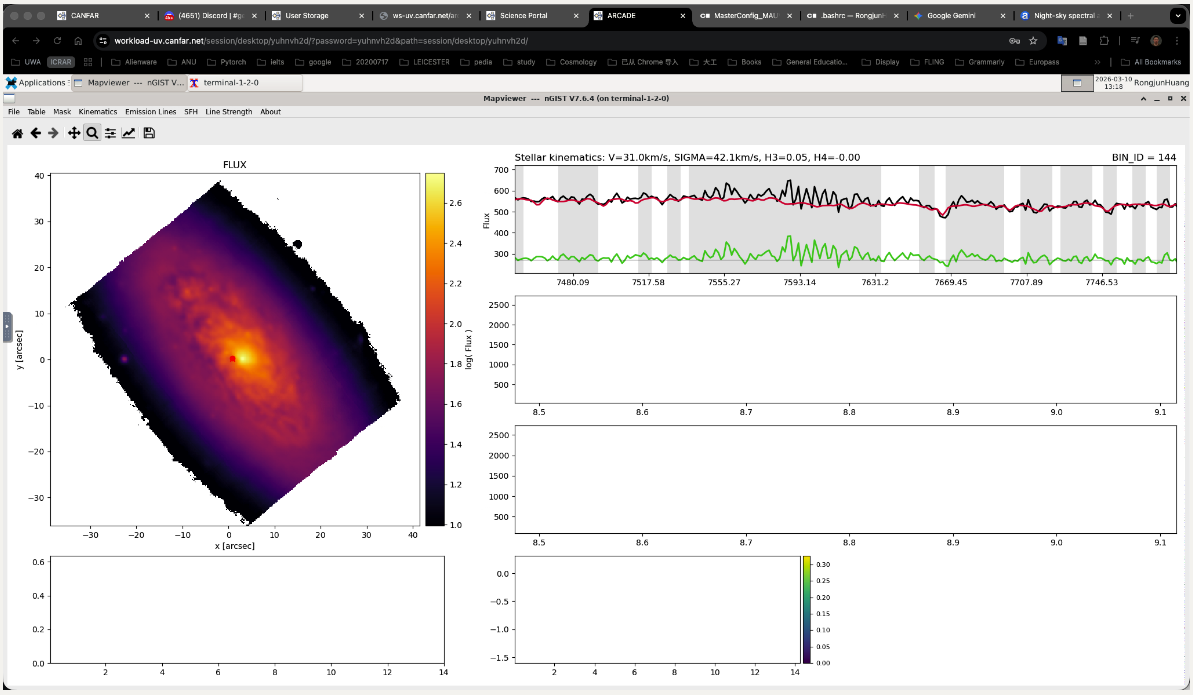
$7000 \sim 7150 Å$ (Argon line the bump before it):



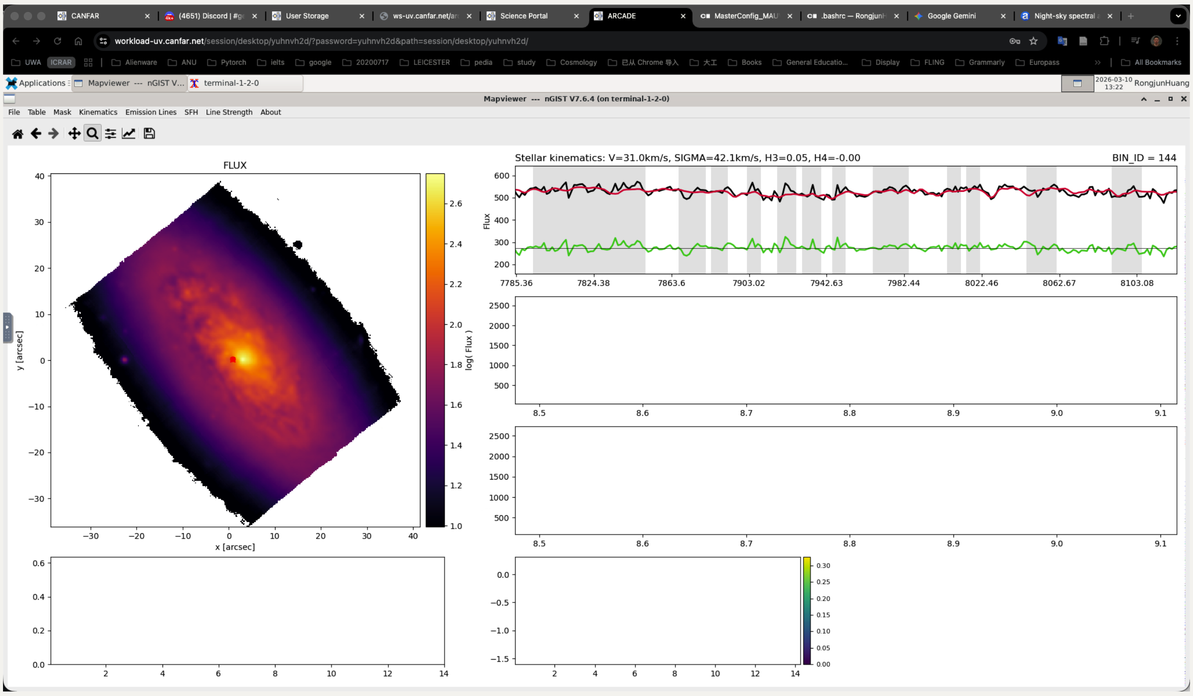
$7150 \sim 7500 Å$ (added noisy sky $7200 - 7240 Å$ region and [O II] lines):



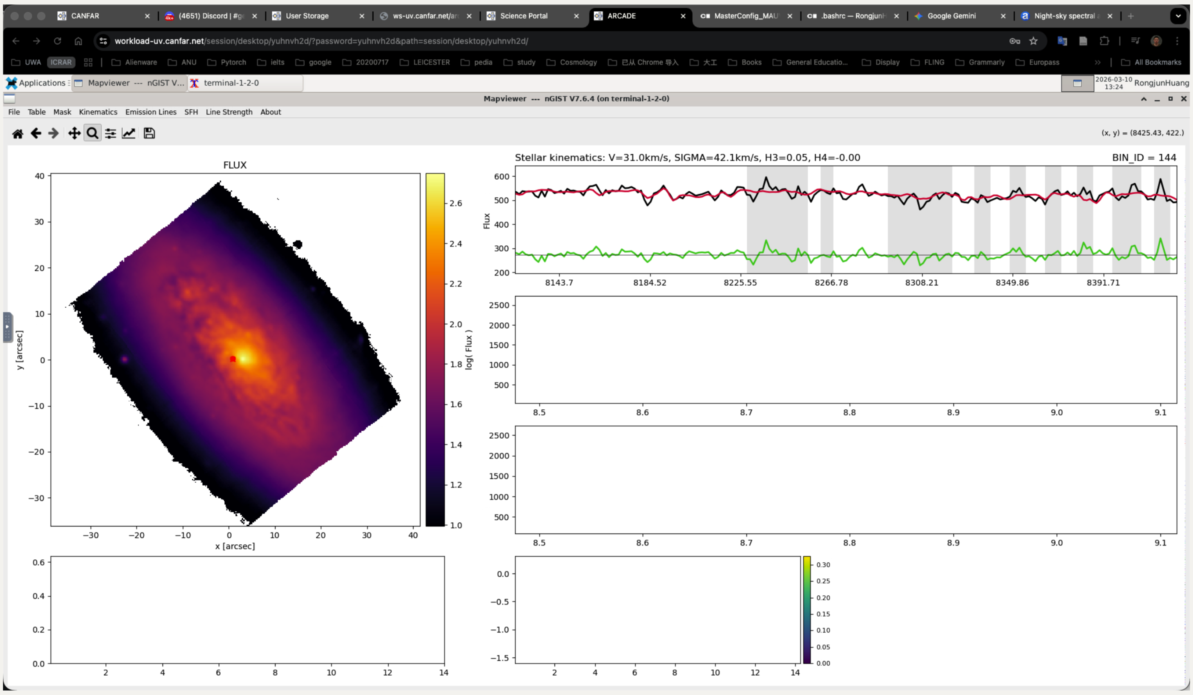
$7500 \sim 7785\AA$ (telluric feature):



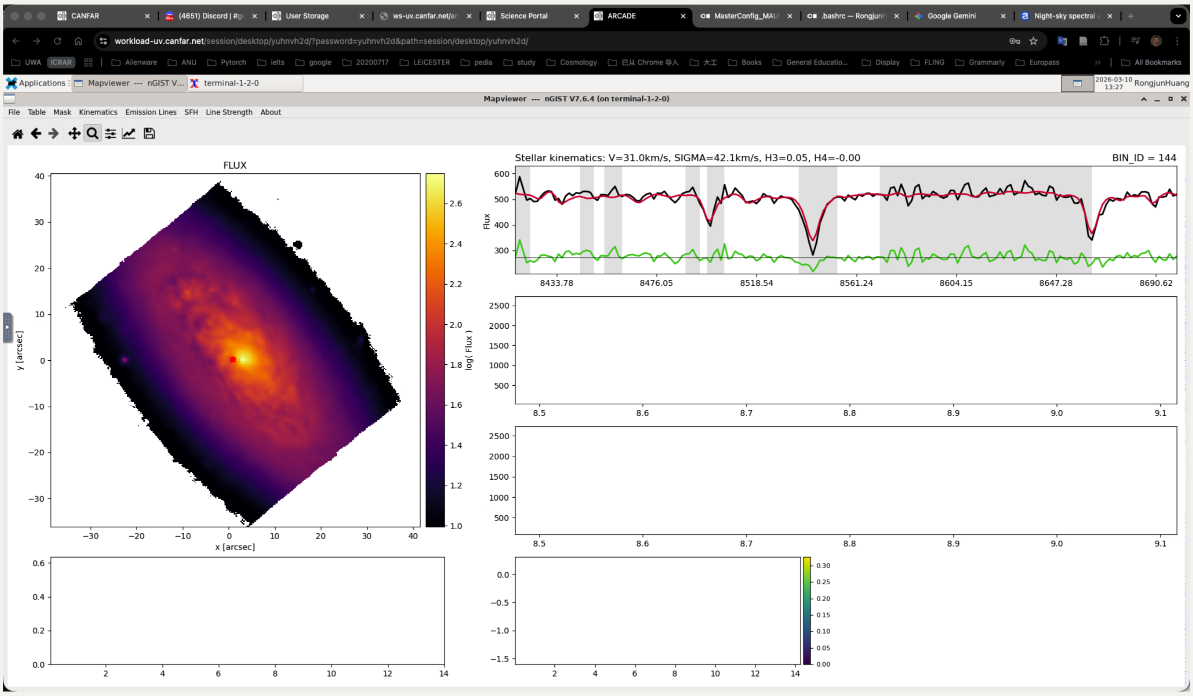
$7785 \sim 8120\AA$:



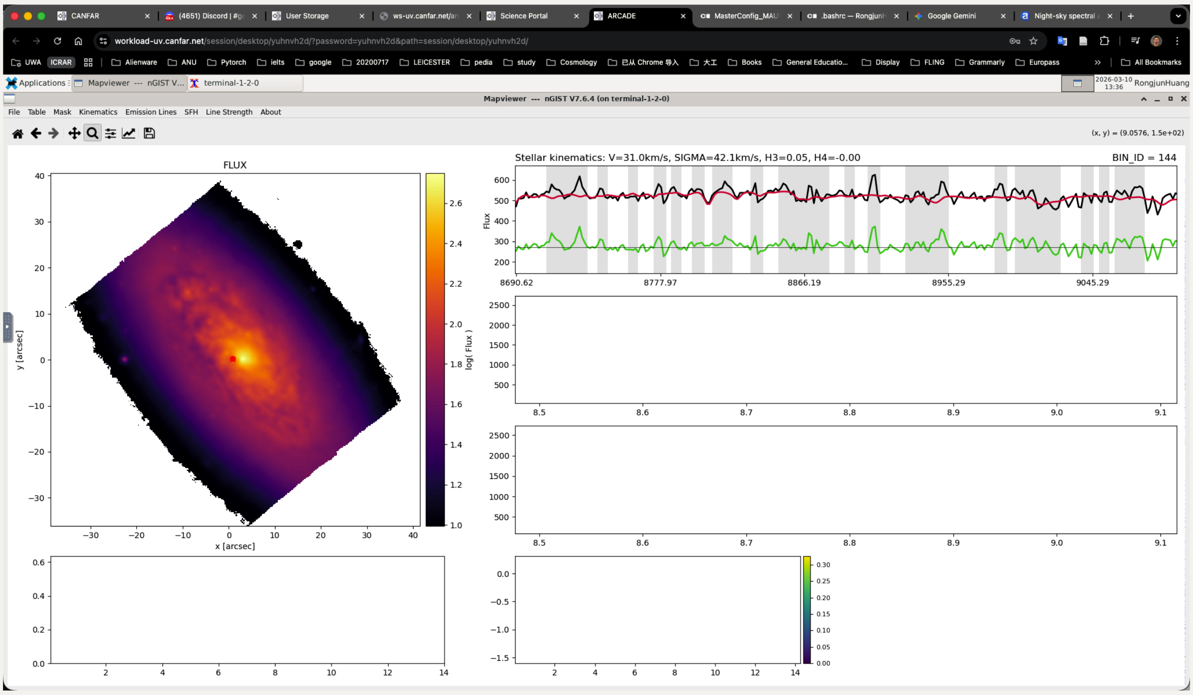
$8120 \sim 8420\AA$:



$8420 \sim 8700\AA$ (CaT absorption features, but dominated by sky emission for IC3392 unfortunately):

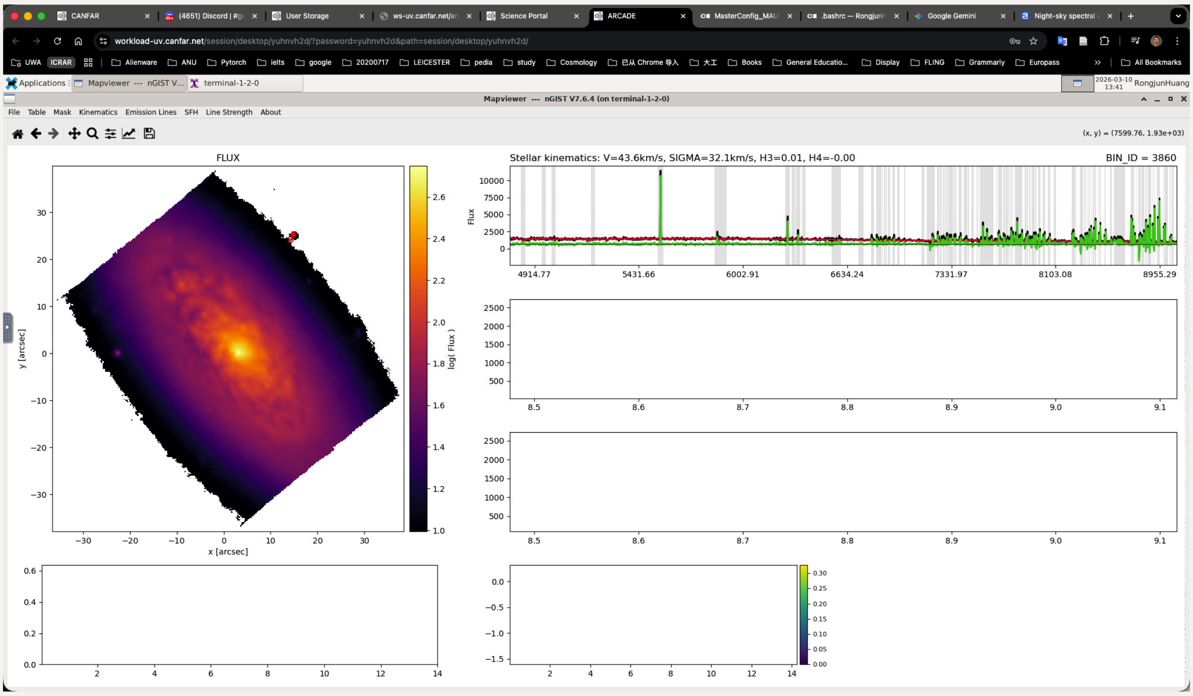


8700 $\sim$ 9100$\mathring{A}$ (Paschen and [S III] lines):

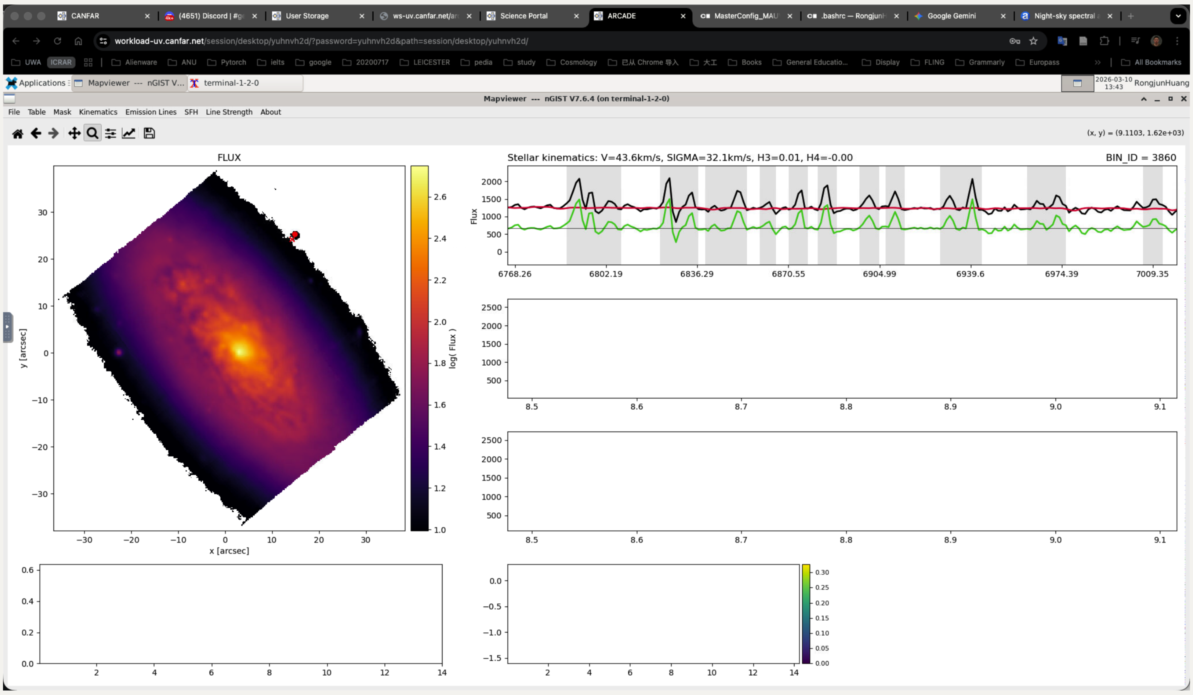


3.2 Faint `Mapviewer`

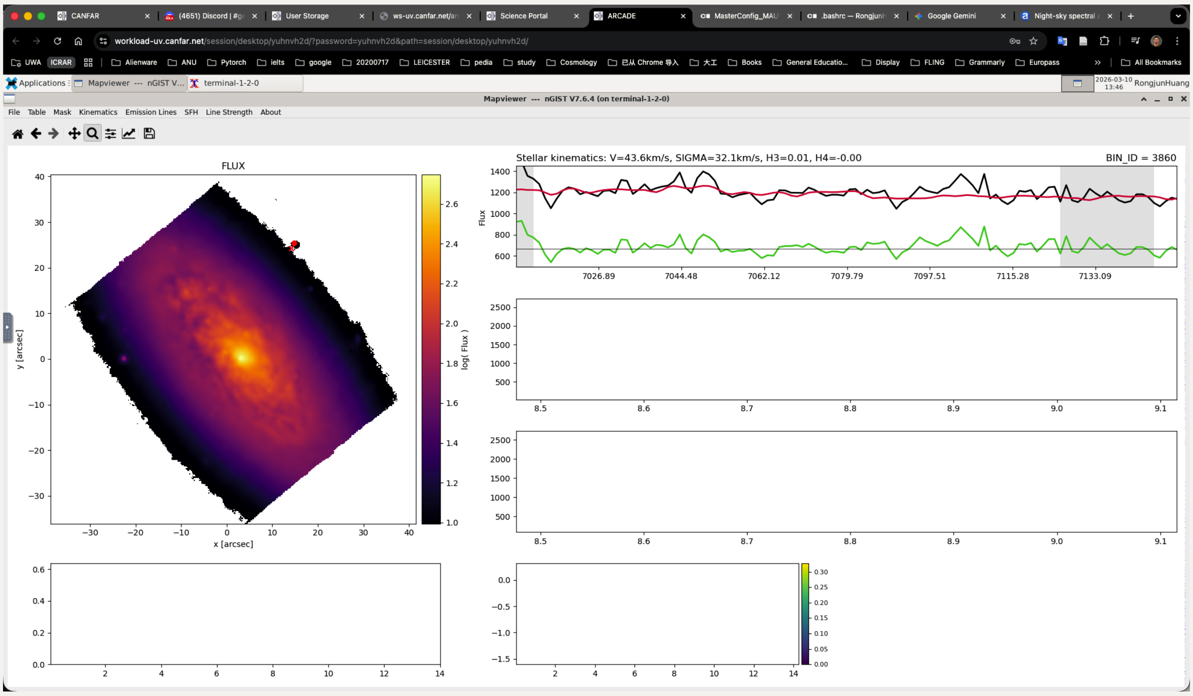
Then, we focus on the isolated upper right bin (still donted by the red dot). Clearly, the spectrum is very noisy but those noisy spikes are masked.



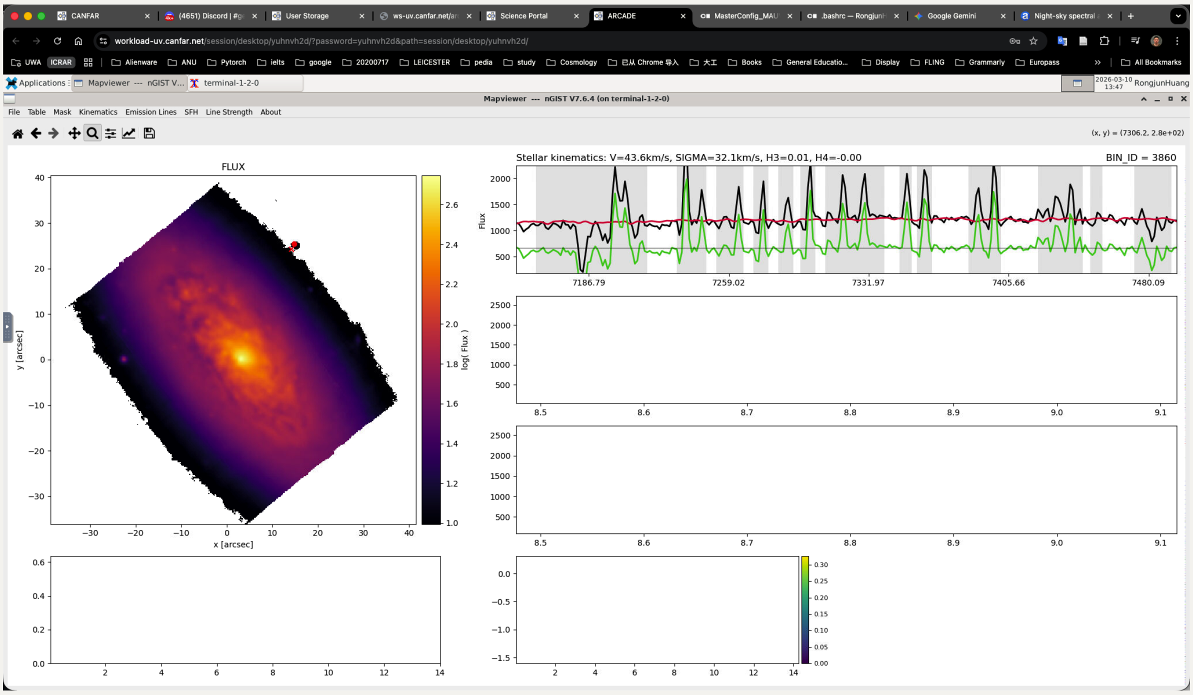
Zoom in to see $6800 \sim 7000\mathring{A}$:



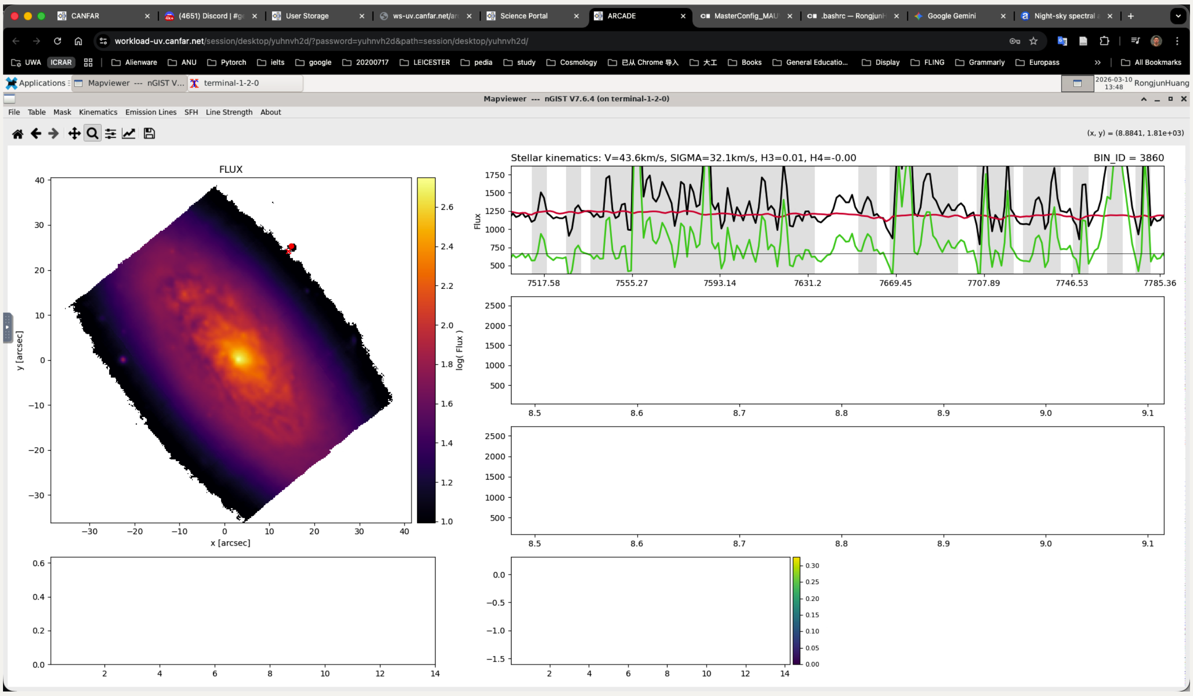
$7000 \sim 7150\mathring{A}$ (Argon line the bump before it):



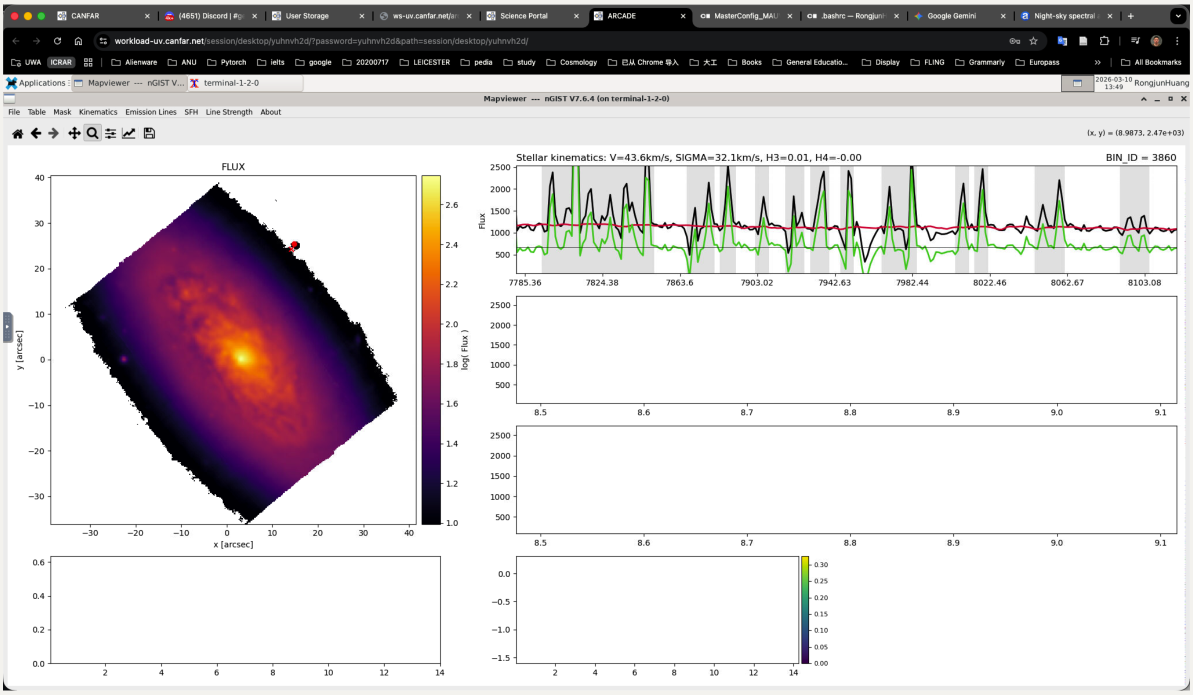
$7150 \sim 7500\mathring{A}$ (added noisy sky $7200 - 7240\mathring{A}$ region and [O II] lines):



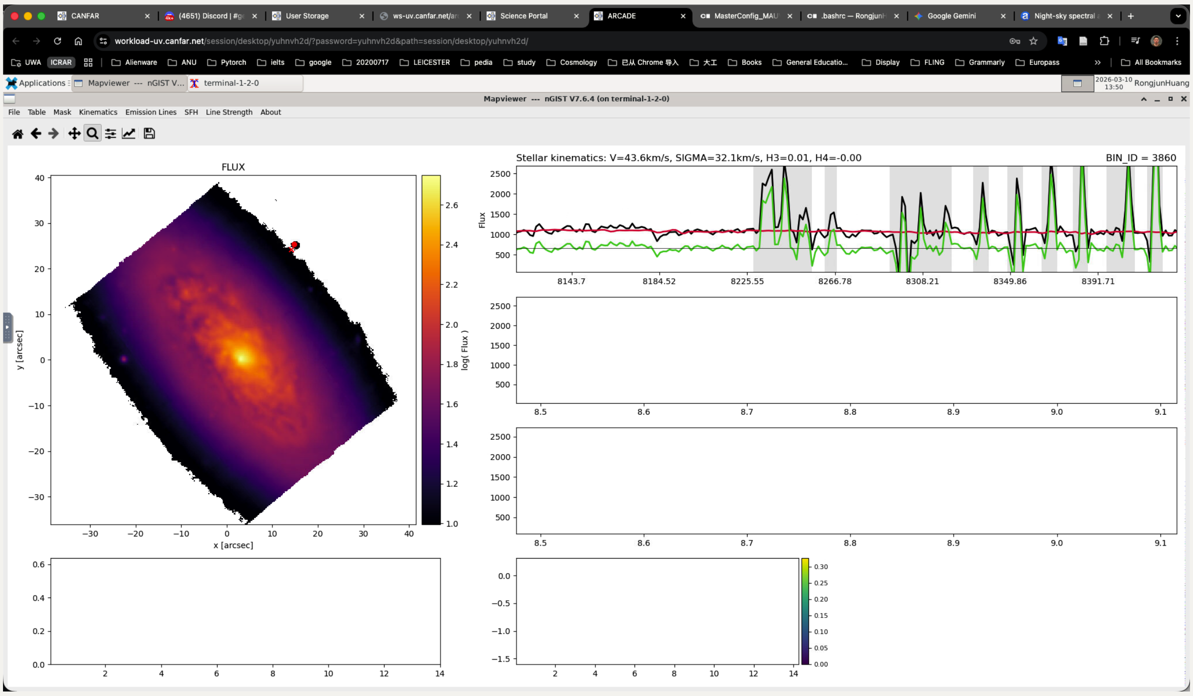
$7500 \sim 7785\mathring{A}$ (telluric feature):



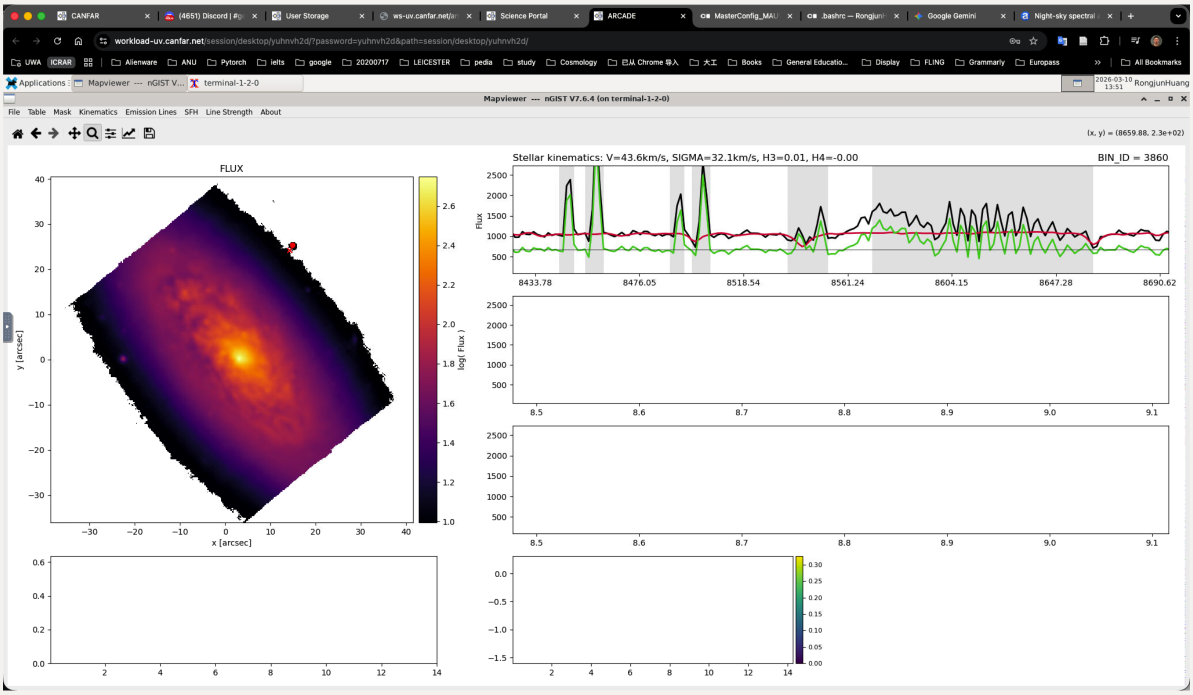
$7785 \sim 8120\mathring{A}$:



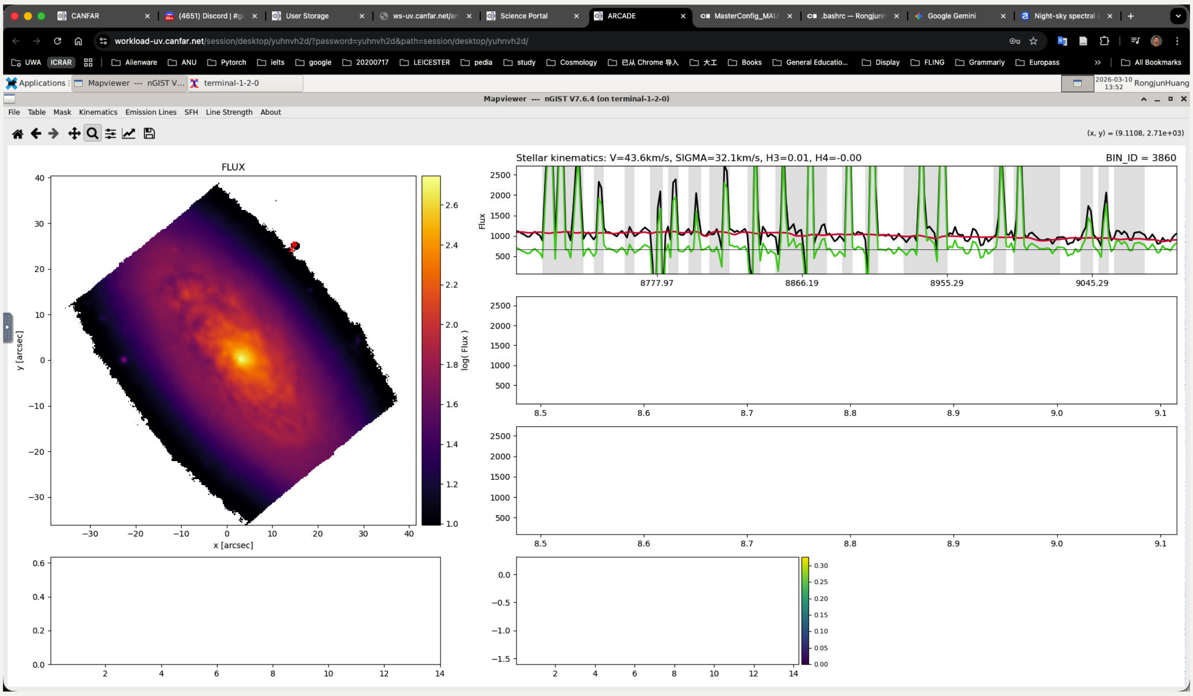
$8120 \sim 8420\mathring{A}$:



$8420 \sim 8700\mathring{A}$ (CaT absorption features, but dominated by sky emission for IC3392 unfortunately):



$8700 \sim 9100\mathring{A}$ (Paschen and [S III] lines):



Mapviewer --- nGIST V7.6.4 (on terminal-1-2-0)
File Table Mask Kinematics Emission Lines SFH Line Strength About
FLUX
x [arcsec]
y [arcsec]
log(Flux)
Stellar kinematics: V=43.6km/s, SIGMA=32.1km/s, H3=0.01, H4=-0.00
BIN_ID = 3860
Flux
8777.97
8866.19
8955.29
9045.29